

California Regional Water Quality Control Board  
North Coast Region

ORDER NO. 89-2  
NPDES PERMIT NO. CA0024627  
I.D. NO. 1A89001NDN

WASTE DISCHARGE REQUIREMENTS

FOR

CALIFORNIA DEPARTMENT OF CORRECTIONS  
PELICAN BAY STATE PRISON

Del Norte County

The California Regional Water Quality Control Board, North Coast Region (hereinafter the Board), finds that:

1. The California Department of Corrections (hereinafter permittee) submitted a Report of Waste Discharge, regarding the Pelican Bay State Prison, dated April 22, 1987. On October 3, 1988 the Department applied for a permit to discharge treated wastewater under the National Pollutant Discharge Elimination System (NPDES).
2. The wastewater treatment and disposal facilities are designed for an average dry weather flow of 0.75 million gallons per day (mgd) and a maximum daily flow of 1.13 mgd. These design flows are based on estimated wastewater flows for a maximum expected occupancy of the prison of 4,332 inmates (2,280-cell prison facility at a crowding factor of 190 percent), prison industries and staff.
3. The proposed "advanced" wastewater treatment facility (to provide pathogen-free and nutrient-controlled effluent) will consist of a nitrification-denitrification activated sludge system (oxidation ditch), coagulation-filtration and disinfection system, a de-chlorinating detention basin and effluent disposal to a series of rapid infiltration basins.
4. Sludge disposal will be by soil injection to a dedicated area on the prison site during dry weather. Sludge will be aerobically treated, then stored in two surface-aerated facultative lagoons during time periods when the injection operations are not in effect. The lagoons will have adequate capacity to prevent overtopping, and supernatant liquid and captured rainfall will be returned to the treatment circuit. The 32-acre area dedicated for sludge soil injection will have adequate surface drainage control and will be managed to balance nitrogen input by uptake with grass-crop harvesting.

5. The de-chlorinating basin and 8.4 acres of treated-effluent infiltration basins are constructed on a 26-acre parcel previously used for lily bulb farming, located approximately two miles northeast of the prison and within the 100-year floodplain of the Smith River. The basins are surrounded by berms with a finish height one foot higher than the standard project (1000 year) flood elevation. Geologic and hydrologic investigations of the site reveal that subsurface conditions at the site are suitable for construction, operation and maintenance of the proposed rapid infiltration basins.
6. This facility is a minor discharger as defined by the U.S. Environmental Protection Agency.
7. The Board adopted the Water Quality Control Plan for the North Coast Region on April 28, 1988. The plan was approved by the State Water Resources Control Board on November 15, 1988.
8. The beneficial uses of Smith River include:
  - a. municipal and domestic supply
  - b. agricultural supply
  - c. industrial service supply
  - d. industrial process supply
  - e. groundwater recharge
  - f. freshwater replenishment
  - g. water contact recreation
  - h. non-contact water recreation
  - i. ocean commercial and sport fishing
  - j. warm freshwater habitat
  - k. cold freshwater habitat
  - l. wildlife habitat
  - m. fish migration
  - n. fish spawning
  - o. shellfish harvesting
9. Beneficial uses of areal groundwaters include:
  - a. domestic water supply
  - b. agricultural water supply
10. The Smith River is a component of the California Wild and Scenic Rivers System due to its extraordinary scenic, recreation, fishery, or wildlife values. The reach between the confluence of the Middle and South Forks to the ocean is classified as "recreational". The Public Resources Code requires all state agencies to exercise their powers in a manner consistent with the policy and provisions of the Wild and Scenic Rivers Act. The Smith River is also a component of the Federal Wild and Scenic Rivers System.



11. The Water Quality Control Plan includes water quality objectives which are not to be exceeded by controllable water quality factors. The Plan contains, as a "General Objective", a reiteration of the State Board's 1968 Statement of Policy with Respect to Maintaining High Quality of Waters in California, Resolution No. 68-16 (the Nondegradation Policy). Pursuant to that policy, whenever existing water quality is better than the numerical objectives established in the Plan, such existing high quality shall be maintained. To implement that Policy, wastewater discharges should be regulated to result in best practicable treatment or controls to assure that existing high quality conditions are maintained.
12. The area of the proposed rapid infiltration basins is underlain by groundwater which contains elevated levels of nitrate nitrogen. Elevated nitrate conditions may be attributed to recent agricultural practices and may naturally decline to levels which exist beneath nearby, uncultivated lands. To minimize degradation of the groundwaters, the discharger's engineers have proposed to incorporate nitrogen-removal processes at the treatment plant. Coagulation-filtration processes for pathogen control will also provide advanced levels of nitrogen and phosphorus removal. Additionally, natural in-soil denitrification processes will be enhanced by providing a basin for de-chlorination of effluent before it is released to the infiltration basins. The Board finds that these measures constitute the best practicable treatment and control of wastewater constituents which might otherwise degrade groundwaters beneath the disposal site. These requirements may be reviewed and revised to reflect information which is derived from future groundwater studies.
13. Geohydrological investigations conducted in preparation of the discharger's Report of Waste Discharge reveal that the alluvial materials in the normally-unsaturated zone beneath the proposed infiltration basins is silty sand, that materials in the saturated zone are gravels and sands, and that groundwater flow velocities beneath the proposed infiltration basin site range from 11 to 33 feet per day. The site is subject to variations in groundwater table elevation due to winter-season increases in flow level of the nearby Smith River. Such variations may result in effluent discharge directly to groundwater during some portion of most years. The direction of groundwater flow is westward (toward existing domestic supply wells) at the same time that the receiving zone is saturated, the flow rate is high and to the west, and conditions are least favorable for pathogen die-off.
14. Wastewater discharged to saturated alluvial materials in proximity to established domestic water supply withdrawals will be "essentially pathogen-free" (as defined in DOHS Draft Policy Statement for Wastewater Reclamation Plants with Direct Filtration). To avoid bacterial or viral degradation of groundwater influenced by the discharge, the discharger will provide advanced pathogen-control processes (coagulation-filtration) before wastewater is released to the basins. A treated-effluent de-chlorination basin and storage basin will provide storage capacity and increase system reliability.

15. Compliance with the Effluent Limitations and Provisions of the proposed Order will allow minimal degradation -- an increase in the Total Dissolved Solids from about 160 mg/l to about 200 mg/l -- of ambient groundwater quality. Such change will be consistent with maximum benefit to the people of the State because of the urgent need to alleviate overcrowded conditions in state prisons. The limited degradation will not unreasonably affect present and anticipated future beneficial uses of the groundwater and will not result in water quality less than that prescribed in the Basin Plan. The water quality and beneficial uses of Smith River will not be affected by the discharge.
16. Effluent limitations, and toxic and pretreated effluent standards established pursuant to Sections 208(b), 301, 302, 303(d), 304, 306, 307, and 403 of the Clean Water Act and amendments thereto are applicable to the discharge.
17. The action to adopt an NPDES permit for Pelican Bay State Prison is exempt from the provisions of the California Environmental Quality Act in accordance with the California Penal Code. An environmental assessment study for the prison, including its wastewater aspects, was completed in November 1986. The Regional Board finds this facility will not cause water quality related effects if conducted in conformance with this Order.
18. On June 12, 1987 the Regional Board adopted Waste Discharge Requirements Order No. 87-49, for the California Department of Corrections (CDC), Pelican Bay State Prison. A petition, requesting review of the Regional Board's action by the State Water Resources Control Board (SWRCB), was filed by the "Concerned Citizens of Del Norte County, an unincorporated association and Richard Mize, M.D." (Petitioners). The petition contended that discharge of waste as regulated by Order No. 87-49 would degrade the Smith River. The SWRCB decided that the petition failed to raise substantial issues that were appropriate for review by the State Board. The Petitioners then filed suit against the SWRCB, the Board, and CDC in the California Superior Court, City and County of San Francisco. The suit petitioned the court for a Writ of Mandate directing the Board to set aside Order No. 87-49 and for related relief against the SWRCB and CDC. The Petitioner's contentions and the Court's decision, as issued on September 1, 1988, are summarized as follows:
  - (A) Contention: The proposed disposal plan violates federal antidegradation requirements. Conclusion: The Regional Board's Order does not violate antidegradation policy.
  - (B) Contention: The proposed plan violates the federal Clean Water Act because it was issued without a National Pollutant Discharge Elimination System (NPDES) Permit. Conclusion: Since the plan does involve the discharge of some quantity of pollutants into the Smith River, albeit indirectly, federal law requires that the plan first be approved by the Environmental Protection Agency through the mechanism of the NPDES permit procedure.



- (C) Contention: The proposed waste disposal authorized by the permit disregards the prohibitions contained in the Water Quality Control Plan adopted for the Klamath River Basin. Conclusion: The manner in which the effluent will be treated and released does not constitute a waste discharge into the Smith River which is absolutely prohibited by the Basin Plan.

The Court also found that there was no basis for the issuance of a writ directed to the SWRCB or CDC. A writ of mandate was issued against the Regional Board directing it to set aside Order No. 87-49 until such time as an NPDES Permit for the proposed waste disposal system has been issued in conformity with law. In response to the Court decision the Regional Board filed a return to the writ of mandate which described the rescission of Order No. 87-49 on September 28, 1988.

The issuance of an NPDES Permit for this project is unique to this situation. Ordinarily, NPDES Permits are not issued for discharges of wastes to land disposal systems. In this case, the Court determined that the rapid infiltration basins within the Smith River floodplain and immediately adjacent to the Smith River require an NPDES Permit. The Regional Board is complying with the Court's order.

19. The Board has notified the permittee and interested agencies and persons of its intent to prescribe waste discharge requirements and issue an NPDES permit for the discharge and has provided them with an opportunity to submit their written comments and recommendations.
20. This Order will serve as a National Pollutant Discharge Elimination System permit pursuant to Section 402 of the Clean Water Act, or amendments thereto, and will take effect at the end of ten days from the date of adoption by the Board.
21. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

THEREFORE, IT IS HEREBY ORDERED that the permittee, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, and the provisions of the Clean Water Act and regulations and guidelines adopted thereunder, shall comply with the following:

A. DISCHARGE PROHIBITIONS

1. The discharge of waste to land that is not under the control of the permittee is prohibited.
2. The discharge of any radiological, chemical, or biological warfare agent or high level radiological waste is prohibited.
3. The discharge of any waste not specifically regulated by this permit is prohibited.

4. The discharge of sludge or digester supernatant not specifically regulated by this permit is prohibited.
5. The discharge of untreated waste from anywhere within the collection, treatment, or disposal facility is prohibited.
6. The discharge of waste from the Pelican Bay State Prison wastewater treatment plant to the Smith River or its tributaries at any time is prohibited.

#### B. EFFLUENT LIMITATIONS

1. Representative sample of the discharge shall not contain constituents in excess of the following limits:

| <u>Constituent</u>            | <u>Unit</u>         | <u>30-Day<br/>Average<sup>a</sup></u>  | <u>7-Day<br/>Average<sup>b</sup></u> | <u>Daily<br/>Maximum</u> |
|-------------------------------|---------------------|----------------------------------------|--------------------------------------|--------------------------|
| BOD (20°C, 5-day)             | mg/l                | 10                                     | 15                                   | 30                       |
|                               | lb/day <sup>c</sup> | 63                                     | 94                                   | 283                      |
| Suspended Solids              | mg/l                | 10                                     | 15                                   | 30                       |
|                               | lb/day <sup>c</sup> | 63                                     | 94                                   | 283                      |
| Settleable Solids             | ml/l                | 0.1                                    | -----                                | 0.1                      |
| Coliform<br>Organisms (Total) | MPN/100 ml          | 2.2 <sup>d</sup>                       | -----                                | 23                       |
| Chlorine Residual             | mg/l                | -----                                  | -----                                | 0.1                      |
| Hydrogen Ion                  | pH                  | Not less than 6.5 nor greater than 8.5 |                                      |                          |
| Grease and Oil                | mg/l                | 15                                     | -----                                | 20                       |
| Nitrate Nitrogen              | mg/l                | 5                                      | -----                                | -----                    |
| Total Phosphorous             | mg/l                | 2                                      | -----                                | 3                        |
| Turbidity                     | NTU                 | 2                                      | -----                                | 5                        |

<sup>a</sup> The arithmetic mean of the values for effluent samples collected in a period of 30 consecutive days.

<sup>b</sup> The arithmetic mean of the values for effluent samples collected in a period of seven consecutive days.



c The daily discharge (lbs/day) is obtained from the following calculation for any calendar day:

$$\text{Daily Discharge (lb/day)} = \frac{8.34}{N} \sum_{i=1}^N Q_i C_i$$

in which N is the number of samples analyzed in any calendar day.  $Q_i$  and  $C_i$  are the flow rate (mgd) and the constituent concentration (mg/l), respectively, which are associated with each of the N grab samples which may be taken in any calendar day. If a composite sample is taken,  $C_i$  is the concentration measured in the composite sample; and  $Q_i$  is the average flow rate occurring during the period over which samples are composited.

d median

2. The arithmetic mean of the BOD (20 C, 5-day) and Suspended Solids values by weight for effluent samples collected in a period of 30 consecutive days shall not exceed 15 percent of the arithmetic mean of the values, by weight, for influent samples collected at approximately the same times during the same period (85 percent removal).
3. The mean daily dry weather flow of waste shall not exceed 0.75 mgd averaged over a period of 30 consecutive days. The peak flow shall not exceed 1.13 mgd.

#### C. RECEIVING WATER LIMITATIONS

1. The discharge shall not contain taste or odor producing substances in concentrations that cause nuisance or adversely affect beneficial uses of the receiving waters.
2. The discharge shall not cause the pH of the groundwater to be depressed below 6.5 nor raised above 8.5.
3. The discharge shall not cause affected groundwaters to exceed the objectives declared in the Basin Plan, nor shall such groundwaters contain concentrations of bacterial indicators or chemical constituents in excess of the limits specified in the California Code of Regulations Title 14 - Drinking Water Standards.
4. The discharge of waste shall not cause esthetically undesirable discoloration of the receiving waters.
5. The discharge shall not contain concentrations of biostimulants which promote objectional aquatic growths to the extent that such growths cause nuisance or adversely affect beneficial uses.
6. The discharge shall not cause groundwater to contain toxic substances in concentrations that are toxic to, or that produce detrimental physiological responses in human, plant or animal life or adversely affect beneficial uses of groundwater.

7. The discharge shall not cause a measurable temperature change.
8. The discharge shall not cause accumulation of pesticide, fungicide or wood treatment chemical concentrations in soils.
9. The discharge shall not cause a visible film or coating on the surface of receiving waters that cause nuisance or that otherwise adversely affect beneficial uses.
10. The discharge shall not cause a violation of any applicable water quality standard adopted by the Board or the State Water Resources Control Board as required by the Federal Water Pollution Control Act, and regulations adopted thereunder. If more stringent applicable water quality standards are promulgated or approved pursuant to Section 303 of the Federal Water Pollution Control Act, or amendments thereto, the Board will revise and modify this permit in accordance with such more stringent standards.

#### D. SOLIDS DISPOSAL

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of at a legal point of disposal, and in accordance with the provisions of Title 23, Chapter 3, Subchapter 15 of the California Code of Regulations. For the purpose of this requirement, the legal point of disposal for treated sludges is defined as the 32-acre site within the prison grounds which is managed in accordance with this Order. Gross solids which are removed from the wastewater treatment train by screening shall be managed as infectious waste in accordance with the Health and Safety Code and regulations promulgated by the Department of Health Services.

#### F. PROVISIONS

##### 1. Duty to Comply

The permittee must comply with all of the conditions of this permit. Any permit noncompliance constitutes a violation of the Clean Water Act and the Porter-Cologne Water Quality Control Act and is grounds for enforcement action; for permit termination, revocation and reissuance, or modification; or denial of a permit renewal application. [40CFR122.41(a)]

The permittee shall comply with effluent standards or prohibitions established under Section 307(a) of the Clean Water Act for toxic pollutants within the time provided in the regulations that establish these standards or prohibitions, even if this permit has not yet been modified to incorporate the requirement. [40CFR122.41(a)(1)]

##### 2. Duty to Reapply

This permit expires on January 26, 1994. If the permittee wishes to continue an activity regulated by this permit after the expiration date of this permit, the permittee must apply for and obtain a new permit. The application, including a report of waste discharge in accordance with Title 23, California Code of Regulations must be received by the Board no later than July 26, 1993. [40CFR122.41(b)]



The Regional Administrator of the Environmental Protection Agency may grant permission to submit an application at a later date prior to the permit expiration date; and the Regional Administrator of the Environmental Protection Agency may grant permission to submit the information required by paragraphs(g)(7), (9), and (10) of 40CFR122.21 after the permit expiration date. [40CFR122.21(d)(2)].

3. Need to Halt or Reduce Activity Not a Defense

It shall not be a defense for a permittee in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this permit. [40CFR122.41(c)]

4. Duty to Mitigate

The permittee shall take all reasonable steps to minimize or prevent any discharge in violation of this permit which has a reasonable likelihood of adversely affecting human health or the environment. [40CFR122.41(d)]

5. Proper Operation and Maintenance

The permittee shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) that are installed or used by the permittee to achieve compliance with this permit. Proper operation and maintenance includes adequate laboratory control and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by a permittee when necessary to achieve compliance with the conditions of this permit. [40CFR122.41(e)]

6. Permit Actions

This permit may be modified, revoked and reissued, or terminated for cause including, but not limited to, the following:

- a. Violation of any terms or conditions of this permit; or
- b. Obtaining this permit by misrepresentation or failure to disclose fully all relevant facts; or
- c. A change in any condition that requires either a temporary or a permanent reduction or elimination of the authorized discharge; or
- d. A determination that the permitted activity endangers human health or the environment and can only be regulated to acceptable levels by permit modification or termination.

The Board may also review and revise this permit at any time upon application of any person, or on the Board's own motion. [CWC 13263(e)]

If any toxic effluent standard or prohibition (including any schedule of compliance specified in such effluent standard or prohibition) is promulgated under Section 307(a) of the Clean Water Act for a toxic pollutant which is present in the discharge and that standard or prohibition is more stringent than any limitation on the pollutant to this permit, this permit shall be modified or revoked and reissued to conform to the toxic effluent standard or prohibition and the permittee so notified. [40CFR122.41(f)]

The filing of a request by the permittee for a permit modification, revocation and reissuance, or termination, or a notification of planned changes or anticipated noncompliance, does not stay any permit condition. [40CFR122.41(f)]

#### 7. Property Rights

This permit does not convey any property rights of any sort, or any exclusive privileges, nor does it authorize any injury to private property or any invasion of personal rights, nor any infringement of federal, state or local laws or regulations. [40CFR122.41(g)]

#### 8. Duty to Provide Information

The permittee shall furnish the Board, State Water Resources Control Board (SWRCB), or Environmental Protection Agency (EPA), within a reasonable time, any information which the Board, SWRCB, or EPA may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this permit or to determine compliance with this permit. The permittee shall also furnish to the Board, upon request, copies of records required to be kept by this permit. [40CFR122.41(h)]

The permittee shall conduct analysis on any sample provided by EPA as part of the Discharge Monitoring Quality Assurance (DMQA) program. The results of any such analysis shall be submitted to EPA's DMQA manager.

#### 9. Inspection and Entry

The permittee shall allow the Board, SWRCB, EPA, and/or other authorized representatives upon the presentation of credentials and other documents as may be required by law, to:

- a. Enter upon the permittee's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this permit;
- b. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this permit;
- c. Inspect at reasonable times any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this permit; and
- d. Sample or monitor at reasonable times, for the purposes of assuring permit compliance or as otherwise authorized by the Clean Water Act, any substances or parameters at any locations. [40CFR122.41(i)]



## 10. Monitoring and Records

- a. Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity.
- b. The permittee shall retain records of all monitoring information, including all calibration and maintenance records and all original strip chart recordings for continuous monitoring instrumentation, copies of all reports required by this permit, and records of all data used to complete the application for this permit, for the life of the project. All monitoring instruments and devices used by the permittee to fulfill the prescribed monitoring program shall be properly maintained and calibrated as necessary, at least annually to ensure their continued accuracy.
- c. Records of monitoring information shall include:
  - i. The date, exact place, and time of sampling or measurements;
  - ii. The individual(s) who performed the sampling or measurements;
  - iii. The date(s) analyses were performed;
  - iv. The individual(s) who performed the analyses;
  - v. The analytical techniques or methods used; and
  - vi. The results of such analyses.
- d. Monitoring must be conducted according to test procedures under 40 CFR Part 136, unless other test procedures have been specified in this permit. Unless otherwise specified, all metals shall be reported as total metals. Test fish for bioassays and test temperatures shall be specified by the Board. Bioassays shall be performed in accordance with guidelines approved by the Board and the Department of Fish and Game.

## 11. Signatory Requirements

- a. All permit applications, reports, or information submitted to the Regional Board, State Board, and/or EPA shall be signed by either a principal executive officer or ranking elected official. [40CFR122.22(a)]
- b. All reports required by this permit, other information requested by the Board, SWRCB, or EPA, and all permit applications submitted for Group II stormwater discharges under 40CFR122.26(b)(3) shall be signed by a person described in paragraph (a) of the provision, or by a duly authorized representative of that person. A person is a duly authorized representative only if:
  - i. the authorization is made in writing by a person described in paragraph (a) of this provision;

- ii. the authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the company (a duly authorized representative may thus be either a named individual or any individual occupying a named position); and
  - iii. the written authorization is submitted to the Board.  
[40CFR122.22(b)]
- c. Changes to authorization. If an authorization under paragraph (b) of this provision is no longer accurate because a different individual or position has responsibility for the overall operation of the facility, a new authorization satisfying the requirements of paragraph (b) of this provision must be submitted to the Board prior to or together with any reports, information, or applications to be signed by an authorized representative. [40CFR122.22(c)]
- d. Any person signing a document under paragraph (a) or (b) of this provision shall make the following certification:
- "I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted, is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations." [40CFR122.22(d)]

## 12. Reporting Requirements

- a. Planned changes: The permittee shall give notice to the Regional Board as soon as possible of any planned physical alteration or additions to the permitted facility. Notice is required under this provision only when:
- i. The alteration or addition to a permitted facility may meet one of the criteria for determining whether a facility is a new source in 40CFR122.29(b); or
  - ii. The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants which are subject neither to effluent limitations in the permit, nor the notification requirements under Provision 17.a.



- b. Anticipated noncompliance: The permittee will give advance notice to the Board of any planned changes in the permitted facility or activity which may result in noncompliance with permit requirements.
- c. Transfers: This permit is not transferable.
- d. Definitions: The following definitions shall apply unless specified in this permit;
  - i. "Daily discharge" means the discharge of a pollutant measured during a calendar day of any 24-hour period that reasonably represents the calendar day for purposes of sampling. For pollutants with limitations expressed in terms of mass, the "daily discharge" is calculated as the total mass of the pollutant discharged over the sampling day. For pollutants with limitations expressed in other units of measurement, the "daily discharge" shall be the concentrations of the composite sample. When grab samples are used, the "daily discharge" determination of concentration shall be the arithmetic average (weighted by flow value) of all samples collected during the sampling day.
  - ii. "Daily average" discharge limitation means the highest allowable average of "daily discharges" over a calendar month, calculated as the sum of all "daily discharges" measured during a calendar month divided by the number of "daily discharges" measured during that month.
  - iii "Daily Maximum" discharge limitations means the highest allowable "daily discharge" during the calendar month.
- e. Monitoring reports: Monitoring results shall be reported at the intervals specified in the monitoring program. If the permittee monitors any pollutant more frequently than required by this permit, using test procedures approved under 40 CFR Part 136 or as specified in this permit, the results of this monitoring shall be included in the calculation and reporting of the data submitted in the monitoring report.
- f. Compliance schedules: Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule of this permit shall be submitted no later than 14 days following each schedule date.
- g. Noncompliance reporting: The permittee shall report any noncompliance that may endanger health or the environment. Any information shall be provided orally as soon as the permittee becomes aware of the circumstances. A written submission shall also be provided within five days of the time the permittee becomes aware of the circumstances. The written submission shall contain a description of the noncompliance and its cause; the period of noncompliance, including exact dates and times and, if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate and prevent recurrence of the noncompliance.

The following shall be included as information that must be reported under this paragraph:

- i. Any unanticipated bypass that exceeds any effluent limitation in the permit.
- ii. Any upset that exceeds any effluent limitation in the permit.
- iii. Violation of a maximum daily discharge limitation for any of the pollutants listed by the Board in this permit.

The Executive Officer may waive the above-required written report.

- h. Other noncompliance: The permittee shall report all instances of noncompliance not reported under paragraphs (a), (e), (f), and (g) of the provision.
- i. Other information: Where the permittee becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Board, the permittee shall promptly submit such facts or information. [40CFR122.41(1)]

### 13. Bypass

The intentional diversion of waste streams from any portion of a treatment facility is prohibited.

### 14. Enforcement

The Clean Water Act provides that any person who violates a permit condition implementing Sections 301, 302, 306, 307, 308, 318, or 405 of the Clean Water Act is subject to a civil penalty not to exceed \$25,000 per day of violation. Any person who negligently violates permit conditions implementing Sections 301, 302, 306, 307, or 308 of the Act is subject to a fine of not less than \$2,500 nor more than \$25,000 per day of violation, or by imprisonment of not more than one year, or both. Higher penalties may be imposed for knowing violations and for repeat offenders. The Porter-Cologne Water Quality Control Act provides for civil and criminal penalties comparable to, and in some cases greater than, those provided under the Clean Water Act.

### 15. General Prohibition

Neither the treatment nor the discharge of waste shall create a pollution, contamination, or nuisance, as defined by Section 13050 of the California Water Code (CWC). [Health and Safety Code, Section 5411]

### 16. Availability

A copy of this permit shall be maintained at the discharge facility and be available at all times to operating personnel.



## 17. Change in Discharge

In the event of a material change in the character, location, or volume of a discharge, (including any point or nonpoint discharge to land or groundwater) the permittee shall file with this Board a new report of waste discharge. [CWC Section 13376]. A material change includes, but is not limited to, the following:

- a. Addition of a major industrial waste discharge to a discharge of essentially domestic sewage, or the addition of a new process or product by an industrial facility resulting in a change in the character of the waste.
- b. Significant change in disposal method, e.g., change from a land disposal to a direct discharge to water, or change in the method of treatment which would significantly alter the characteristics of the waste.
- c. Significant change in the disposal area, e.g., moving the discharge to another drainage area, to a different water body, or to a disposal area, significantly removed from the original area, potentially causing different water quality or nuisance problems.
- d. Increase in area or depth to be used for solid waste disposal beyond that specified in the waste discharge requirements. [CCR Title 23 Section 2210]

## 18. Vested Rights

This permit does not convey any property rights of any sort or any exclusive privileges. The requirements prescribed herein do not authorize the commission of any act causing injury to persons or property, nor protect the permittee from his liability under Federal, State or local laws, nor create a vested right for the permittee to continue the waste discharge.

## 19. Severability

Provisions of these waste discharge requirements are severable. If any provision of these requirements is found invalid, the remainder of these requirements shall not be affected.

## 20. Monitoring

The permittee must comply with the Contingency Planning and Notification Requirements Order No. 74-151 and the Monitoring and Reporting Program No. 89-2 and any modifications to these documents as

specified by the Executive Officer. Such documents are attached to this permit and incorporated herein. The permittee shall file with the Board technical reports on self monitoring work performed according to the detailed specifications contained in any monitoring and reporting program as directed by the Board.

Chemical, bacteriological, and bioassay analyses shall be conducted at a laboratory certified for such analyses by the State Department of Health Services. In the event a certified laboratory is not available to the permittee, analyses performed by a noncertified laboratory will be accepted provided:

- a. A quality assurance/quality control program is instituted by the laboratory. A manual containing the steps followed in this program must be kept in the laboratory and shall be available for inspection by staff of the Board. The quality assurance/quality control program must conform to EPA guidelines or procedures approved by the Board.
- b. The laboratory will become certified within the shortest practicable time if the State certification program is resumed.

#### 21. Operator Certification

Supervisors and operators of municipal wastewater treatment plants shall possess a certificate of appropriate grade in accordance with Title 23, California Code of Regulations, Section 3680. The State Board may accept experience in lieu of qualification training. In lieu of a properly certified wastewater treatment plant operator, the State Board may approve use of a water treatment plant operator of appropriate grade certified by the State Department of Health Services where reclamation is involved.

#### 22. Warfare Agents

The discharge of any radiological, chemical, or biological warfare agent to waters of the State is prohibited. [CWC Section 13275]

#### Certification

I, Benjamin D. Kor, Executive Officer, do hereby certify that the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, North Coast Region, on January 26, 1989.

**Original Signed By:**

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Benjamin D. Kor  
Executive Officer



California Regional Water Quality Control Board  
North Coast Region

MONITORING AND REPORTING PROGRAM NO. 89-2  
FOR

CALIFORNIA DEPARTMENT OF CORRECTIONS  
PELICAN BAY STATE PRISON

Del Norte County

MONITORING

WASTEWATER TREATMENT FACILITY:

The purposes of monitoring the wastewater treatment facility are:

- To ensure adequate treatment of the wastewater including removal of suspended and floatable material, treatment of biodegradable organics, and elimination of pathogenic organisms.
- To routinely sample and analyze the treatment process; and
- To produce a sequence of reports which document the discharger's consistent compliance regarding adequate treatment, operation and maintenance.

Composite samples may be taken by a proportional sampling device approved by the Executive Officer or by grab samples composited in proportion to flow. In compositing grab samples, the sampling interval shall not exceed one hour. The following shall constitute the monitoring program:

Influent Monitoring

| <u>Constituent</u>    | <u>Units</u> | <u>Type of Sample</u> | <u>Frequency</u> |
|-----------------------|--------------|-----------------------|------------------|
| BOD (20°C, 5-day)     | mg/l         | 8-hour composite      | twice weekly     |
| Total Phosphorous     | mg/l         | 8-hour composite      | monthly          |
| Nonfilterable Residue | mg/l         | 8-hour composite      | daily            |
| Flow (Mean and Peak)  | mgd          | continuous            | daily            |

Effluent Monitoring

Samples are to be taken after disinfection.

| <u>Constituent</u> | <u>Units</u> | <u>Type of Sample</u> | <u>Frequency</u> |
|--------------------|--------------|-----------------------|------------------|
| Mean Daily Flow    | mgd          | continuous            | daily            |
| BOD (20°C, 5-day)  | mg/l         | 8-hour composite      | twice weekly     |
| Settleable Matter  | ml/l         | grab                  | daily            |

| <u>Constituent</u>                             | <u>Units</u> | <u>Type of Sample</u> | <u>Frequency</u> |
|------------------------------------------------|--------------|-----------------------|------------------|
| Nonfilterable Residue                          | mg/l         | 8-hour composite      | daily            |
| Grease and Oil                                 | mg/l         | grab                  | monthly          |
| Hydrogen Ion                                   | pH           | grab                  | daily            |
| Coliform Organisms                             | MPN/100 ml   | grab                  | daily            |
| Chlorine Residual                              | mg/l         | grab                  | daily            |
| Turbidity                                      | NTU          | continuous            | daily            |
| Nitrate Nitrogen                               | mg/l         | 8-hour composite      | twice weekly     |
| Ammonia Nitrogen                               | mg/l         | 8-hour composite      | twice weekly     |
| Total Kjeldahl Nitrogen                        | mg/l         | 8-hour composite      | twice weekly     |
| Organic Nitrogen                               | mg/l         | 8-hour composite      | twice weekly     |
| Total, Organic and Reactive Phosphorus         | mg/l         | 8-hour composite      | twice weekly     |
| Chloride                                       | mg/l         | 8-hour composite      | twice weekly     |
| Halogenated Volatile Organics (EPA Method 601) | mg/l         | grab                  | quarterly        |

GROUNDWATER MONITORING:

Groundwater monitoring is conducted to:

- Identify water quality parameters which may be used to detect and/or quantify impacts of sludge application practices or treated wastewater discharge to the Rapid Infiltration Basins (RIBs);
- Describe water quality in groundwater prior to and after application of municipal sludge and discharge of treated wastewater to the RIBs;
- Obtain routine sampling and analysis information of groundwater quality conditions;
- Produce a sequence of reports which reveal whether or not impacts to groundwater quality have occurred due to sludge application or discharge of treated wastewater to RIBs.

Pre-Discharge Monitoring

Describe existing groundwater flow gradients. Install and maintain a network of piezometers within and proximate to the sludge disposal and rapid infiltration basins; including at least one "background" monitoring well located in an area unaffected by the commencement of discharge.

Describe groundwater quality conditions. Install and maintain a network of sampling wells within and proximate to the sludge disposal and rapid infiltration basins; sample and analyze the spatial and temporal variations of existing groundwater quality using the following parameters:



| <u>Constituent</u>     | <u>Units</u> | <u>Type of Sample</u> | <u>Frequency</u> |
|------------------------|--------------|-----------------------|------------------|
| Total Dissolved Solids | mg/l         | grab                  | monthly          |
| Nitrate Nitrogen       | mg/l         | grab                  | monthly          |
| Orthophosphate         | mg/l         | grab                  | monthly          |
| Chloride               | mg/l         | grab                  | monthly          |

Propose a network of sampling/monitoring wells to serve as checkpoints for compliance with this Order.

#### Discharge Monitoring

Samples shall be taken from an established network of groundwater monitoring wells representative of conditions upgradient and downgradient of the sludge disposal area and the rapid infiltration basins. The locations and construction details of the monitoring wells shall be in accordance with specifications issued by the Executive Officer.

Samples shall be analyzed for the following:

| <u>Constituent</u>     | <u>Units</u> | <u>Type of Sample</u> | <u>Frequency</u> |
|------------------------|--------------|-----------------------|------------------|
| Total Dissolved Solids | mg/l         | grab                  | monthly          |
| Nitrate Nitrogen       | mg/l         | grab                  | monthly          |
| Orthophosphate         | mg/l         | grab                  | monthly          |
| Chloride               | mg/l         | grab                  | monthly          |

Continue to maintain the established network of piezometers; observe, record and analyze the variations in piezometric surface; compare and analyze data obtained prior to discharge with data obtained after commencement of discharge.

#### SLUDGE DISPOSAL AREA:

##### Pre-discharge Monitoring

A representative sample of soil from the sludge application area shall be collected and analyzed for metals (ICAP Scan).

##### Discharge Monitoring

Annually, following the completion of sludge disposal for the year, the discharger shall provide a report to the Board which shall include but not be limited to the following:

- The quantity of sludge disposed (in dry tons and gallons);
- The quality of sludge disposed (ie., well stabilized, odor free) and concentrations of nitrogen and the metals identified and quantified from the quarterly samples.
- The specific location on the application site where the sludge was injected or spread.
- Analyses of the soil pH prior to sludge disposal.
- The amount of nitrogen (N) in pounds applied per acre.

To assess the amount of nutrients removed by cropping, the discharger shall analyze a representative sample of the harvested crop for Total Nitrogen and Total Phosphorus. The crop analysis shall be used to assess the relative contribution of nutrients provided by sludge.

This report shall be submitted to the Regional Board and EPA no later than December 31st of each year.

Following the fourth year of sludge disposal, a representative sample of soil shall be collected from the application area and analyzed for the metals encountered during the quarterly sludge analyses. The results shall be included in the above report.

#### SMITH RIVER MONITORING:

The objectives of Smith River monitoring are to:

Describe ambient water quality conditions in the river prior to, and concurrent with the start of treated wastewater discharge to the Rapid Infiltration Basins (RIBs);

Identify water quality parameters which may be used to detect and/or quantify impacts of treated wastewater discharge;

Perform routine sampling and analysis of water quality conditions;

Produce a sequence of reports which reveal whether or not impacts of discharge have been detected.

#### Periodic Video-record Survey:

During the 1989 pre-discharge monitoring period and at subsequent time periods as specified by the Regional Board, the discharger shall obtain low-level aerial color-video tape recordings of the lower Smith River (between the Mouth and the Crescent City water intake caisson). The videograph shall cover the water surface and adjacent gravel bars and be directed to document the presence/absence and/or extent of algae accumulations along the lower eight-mile reach of the river. Commercial-quality (three-cell) high-resolution camera shall be used and VHS-format tapes shall be provided. Still photographs shall be taken at the same time to document the five surface water monitoring stations.

#### Summer-season Surface Water Monitoring:

The following analysis series shall be performed at the stations indicated:

Upper Station. Proximate to the Crescent City Water Intake.

Water Column Series<sup>1</sup>

Periphyton Series<sup>2</sup>

Continuous Multiparameter Survey<sup>3</sup>

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See next page for Footnotes 1, 2, and 3.



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- 1/ Water Column Series consists of analyses of samples taken from well-mixed section of the river (except at the Below Rowdy Creek station, where sample series shall be obtained at two depths: above the pycnocline and below the pycnocline) at the locations indicated and analyzed for the following:

Field Parameters

Temperature  
Electrical Conductivity  
pH  
Dissolved Oxygen

Lab Parameters:

Turbidity  
Nitrate-N  
Total Kjeldahl-N  
Total reactive phosphorus  
Chloride  
Total Iron  
Biomass (chlorophyll a / adenosine triphosphate)

The frequency shall be monthly during the low flow period of the year, June through the beginning of winter runoff.

- 2/ Periphyton Series consists of descriptive (identification to genus and ranking as to relative abundance) and quantitative (dry weight and ash-free dry weight per unit of surface) information derived after deploying and retrieving standard plexiglass substrates as described by the following:

USGS, 1977. Methods for collection and analysis of aquatic biological and microbiological samples. Chapter A4 (eds.: Greeson, et al). Techniques of Water-Resources Investigation of the United State Geological Survey. Book 5. Washington DC.

USGS, 1979. A supplement to: Methods for collection and analysis of aquatic biological and microbiological samples (U. S. Geological Survey Techniques of Water-Resources Investigations, Book 5, Chapter A4). U. S. Geological Survey. Open-File Report 79-1279. Washington DC.

The period of substrate deployment shall bracket the dates of Water Column sampling and the period during which Continuous Multiparameter Survey is conducted. The start of this series may be delayed beyond June if cold temperatures or high flows persist.

- 3/ Continuous Multiparameter Survey consists of sequential recording of Temperature, pH, Electrical Conductivity and Dissolved Oxygen via an in-situ probe which is deployed in the river for at least a 48-hour period. The time of deployment shall include the date that Water Column sampling is done and shall "nest" within the period that Periphyton substrate is deployed.
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Bridge Station. Proximate to Dr. Fine Bridge (Hwy 101)

Water Column  
Periphyton

County Ramp Station. (On S. side may be above Bailey Hole)

Water Column  
Periphyton  
Continuous Multiparameter

Above Rowdy Creek Station. (On S. side via Tedson ranch gravel bar)

Water Column  
Periphyton  
Continuous Multiparameter

Below Rowdy Creek Station. (Above Cattle Crossing)

Water quality assessments at this station must accommodate the variations which occur because of tidal and oceanic influences. Accordingly, the discharger shall employ qualified personnel to obtain samples and data recordings which are representative of the range of conditions encountered (a) between lower-low and higher-high tide levels and (b) between that portion of the water column which is above the pycnocline and the portion which is below the pycnocline.

Monitoring and sampling equipment shall be operated in a manner that concurrently characterizes the "upper band" (low density water, above the pycnocline) and the "lower band" (high density, below the pycnocline) as they ebb and flood past a fixed point.

Water Column (upper and lower)  
Periphyton (lower only)  
Continuous Multiparameter (upper and lower)

#### REPORTING

Monitoring reports shall be submitted to the Board for each month on or before the fifteenth day of the following month. In reporting the monitoring data, the discharger shall arrange the data on a form approved by the Executive Officer. The data shall be summarized in such a manner to illustrate clearly the compliance with waste discharge requirements. Copies of each monitoring report shall be mailed to:

Regional Administrator  
U.S. Environmental Protection Agency  
Region 9  
215 Fremont Street  
San Francisco, CA 94105

Ordered by Original Signed by:  
Benjamin D. Kor  
Executive Officer

January 26, 1989



California Regional Water Quality Control Board  
North Coast Region

CONTINGENCY PLANNING AND NOTIFICATION REQUIREMENTS

FOR

ACCIDENTAL SPILLS AND DISCHARGES

ORDER NO. 74-151

The California Regional Water Quality Control Board, North Coast Region, finds that:

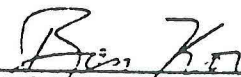
1. Section 13225 of the Porter-Cologne Water Quality Control Act requires the Regional Board to perform general duties to assure positive water quality control.
2. The Regional Board has been advised of situations in which preparations for, and response to accidental discharges and spills have been inadequate.
3. Persons discharging waste or conveying, supplying, storing, or managing wastes or hazardous materials have the primary responsibility for contingency planning, incident reporting and continuous and diligent action to abate the effects of such unintentional or accidental discharge.

THEREFORE, IT IS HEREBY ORDERED THAT:

- I. All persons who discharge wastes or convey, supply, store, or otherwise manage wastes or other hazardous material shall:
  - A. Prepare and submit to this Regional Board, according to a time schedule prescribed by the Executive Officer, a contingency plan defining the following:
    1. Potential locations and/or circumstances under which accidental discharge incidents might be expected to occur,
    2. Possible water quality effects of accidental discharges,
    3. The conceptual plan for cleanup and abatement of accidental discharge incidents, including:
      - a. The individual who will be in charge of cleanup and abatement activities on behalf of the discharger,
      - b. The equipment and manpower available to the discharger to implement the cleanup and abatement plans,
  - B. Immediately report to the Regional Board any accidental discharge incidents. Such notification shall be made by telephone as soon as the responsible person or his agent has knowledge of the incident.

- C. Immediately begin diligent and continuous action to cleanup and abate the effects of any unintentional or accidental discharge. Such action shall include temporary measures to abate the discharge prior to completing permanent repairs to damaged facilities.
  - D. Confirm the telephone notification in writing within two weeks of the telephone notification. The written notification shall include: reasons for the discharge, duration and volume of the discharge, steps taken to correct the problem and steps being taken to prevent the problem from recurring.
- II. Upon original receipt of phone report (I.B), the Executive Officer shall immediately notify all affected agencies and known users of waters affected by the unintentional or accidental discharge.
- III. Provide updated information to the Regional Board in the event of change of staff, size of the facility, or change of operating procedures which will affect the previously established contingency plan.
- IV. The Executive Officer or his employees shall maintain liaison with the discharger and other affected agencies and persons to provide assistance in cleanup and abatement activities.
- V. The Executive Officer shall transmit copies of this Order to all persons whose discharges of waste handling activities are governed by Waste Discharge Requirements or an NPDES Permit. Such transmittal shall include a current listing of telephone numbers of the Executive Officer and his key employees to facilitate compliance with Item I.B of this Order.

Ordered by

  
Benjamin D. Kor  
Executive Officer

July 24, 1974  
(Retyped January, 1986)

Your primary notification should be to the Regional Board office at Santa Rosa at (707) 576-2220. During off hours, you will be able to leave a recorded message at that number and, if you have a spill or discharge emergency, you will also be referred to the State Office of Emergency Services (OES) at (800) 852-7550. OES maintains a roster of key employees and will relay your notification to Regional Board staff.



California Regional Water Quality Control Board  
North Coast Region

GENERAL MONITORING AND REPORTING PROVISIONS

February 28, 1973  
(Revised on April 19, 1983)

GENERAL PROVISIONS FOR SAMPLING AND ANALYSIS

Unless otherwise noted, all sampling and sample preservation, shall be in accordance with the current edition of "Standard Methods for the Examination of Water and Wastewater". All analyses shall be conducted in accordance with the test procedures approved under 40 CFR Part 136 "Test Procedures for the Analysis of Pollutants".

All analyses shall be performed in a laboratory certified to perform such analyses by the California State Department of Health or a laboratory approved by the Executive Officer.

All samples shall be representative of the waste discharge. Effluent samples shall be taken downstream of any addition to the treatment works and prior to mixing with the receiving waters.

The discharger shall calibrate and perform maintenance procedures on all monitoring instruments and equipment to ensure accuracy of measurements.

GENERAL PROVISIONS FOR REPORTING

For every item where the requirements are not met, the discharger shall submit a statement of the actions undertaken or proposed which will bring the discharge into full compliance with requirements at the earliest time and submit a timetable for correction.

By January 30 of each year, the discharger shall submit an annual report to the Regional Board. The report shall contain both tabular and graphical summaries of the monitoring data obtained during the previous year. In addition, the discharger shall discuss the compliance record and the corrective actions taken or planned which may be needed to bring the discharge into full compliance with the waste discharge requirements.

The discharger shall maintain all sampling and analytical results, including strip charts; date, exact place, and time sampling; date analyses were performed; analyst name, analytical techniques used; and result of all analyses. Such records shall be retained for a minimum of three years. This period of retention shall be extended during the course of any unresolved litigation regarding this discharge.

Monitoring reports shall be signed by:

- a. In the case of corporations, by a principal executive officer at least of the level of vice-president or his duly authorized representative, if such representative is responsible for the overall operation of the facility from which the discharge originates.

General Monitoring and  
Reporting Provisions

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- b. In the case of a partnership, by a general partner.
- c. In the case of a sole proprietorship, by the proprietor.
- d. In the case of a municipal, state or other public facility, by either a principal executive officer, ranking elected official, or other authorized employee.